

ABB ACS580 VFD 75kW — Installation and Repair Guide

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EQUIPMENT

Tag: EQ-VFD-ACS580-04 Plant: North Refinery Unit 4
Model: ACS580-01-145A-4 Category: Variable Frequency Drive
Serial: ABB-580-77321
Manufacturer: ABB (Switzerland)

SAFETY HAZARDS

- Capacitor discharge
- Arc flash
- DC bus residual voltage

AUTHORS

- T-1042 Priya Subbu (Electrical)
- T-1290 Henrik Sand (Instrumentation)

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Section 3 — Fuse Replacement

Procedure ID: PRC-VFD-003-A

Category: Corrective Maintenance Severity: High

Performed by: T-1042 Priya Subbu (Electrical)

OVERVIEW

This procedure covers replace input fuses after fault trip on equipment EQ-VFD-ACS580-04 (ACS580-01-145A-4). Follow the safety hazards listed on page 1 before starting. Lock out and tag out upstream isolation before disassembly.

PARTS REQUIRED

- VFD-FUSE-32A — VFD Input Fuse Class J 32A (Electrical)

STEPS

1. Confirm isolation per plant LOTO procedure. Verify zero energy state.
2. Stage the parts listed above plus a calibrated torque wrench and clean rags.
3. Disassemble per Section 3 of this manual.
4. Inspect mating surfaces for wear, scoring, or corrosion. Photograph any anomalies.
5. Install replacement parts in the order listed. Torque fasteners per Table 2.
6. Reassemble and reinstate auxiliaries (lubrication, cooling, instrumentation).
7. Remove isolation and perform a no-load function test.
8. Record completion against work order in CMMS.

ACCEPTANCE CRITERIA

- No abnormal vibration or noise during no-load run.
- All fasteners torqued and witness-marked.
- No leaks at static or dynamic seals after a 5-minute run.

Section 4 — Calibration

Procedure ID: PRC-VFD-003-B

Category: Predictive Maintenance Severity: Low

Performed by: T-1290 Henrik Sand (Instrumentation)

OVERVIEW

This procedure covers calibrate output current measurement on equipment EQ-VFD-ACS580-04 (ACS580-01-145A-4). Follow the safety hazards listed on page 1 before starting. Lock out and tag out upstream isolation before disassembly.

PARTS REQUIRED

- PR-GAUGE-10BAR — Bourdon Pressure Gauge 0-10 bar (Instrumentation)

STEPS

1. Confirm isolation per plant LOTO procedure. Verify zero energy state.
2. Stage the parts listed above plus a calibrated torque wrench and clean rags.
3. Disassemble per Section 4 of this manual.
4. Inspect mating surfaces for wear, scoring, or corrosion. Photograph any anomalies.
5. Install replacement parts in the order listed. Torque fasteners per Table 2.
6. Reassemble and reinstate auxiliaries (lubrication, cooling, instrumentation).
7. Remove isolation and perform a no-load function test.
8. Record completion against work order in CMMS.

ACCEPTANCE CRITERIA

- No abnormal vibration or noise during no-load run.
- All fasteners torqued and witness-marked.
- No leaks at static or dynamic seals after a 5-minute run.